Batch 1-Micro Projects

S.No	Group	Title	
1	NAVANITHAN (URK25CO2013) YOGESH (URK25CO2014) KAMALESHWARAN (URK25CO2015) NAVEEN (URK25CO2016)	**Micro Project - 01**	Network Threat Detection in Cyber Security Using Bayes' Theorem
		Industry Focus	Cybersecurity & Intelligent Systems
		Integrated COs	CO1, CO3, CO5
		Project Scope	This project has strong future scope in real-time cyber threat detection for organizations like Palo Alto Networks, Cisco, and Fortinet. Using Bayes' Theorem enables probabilistic reasoning under uncertainty to classify suspicious network events. The approach supports intelligent decision-making systems for intrusion detection and predictive analytics. It can be extended to autonomous security monitoring platforms used in smart infrastructures.
2.	Bavana URK25AI1151 Thaniska URK25AI1141 Bhavya URK25AI1137 Judith Agnes URK25AI1156	**Micro Project - 02**	Email Spam Filtering Using Bayesian Probability.
		Industry Focus	Enterprise IT & Cloud Services
		Integrated COs	CO1, CO2, CO5
		Project Scope	The project supports intelligent email security solutions used in enterprise and cloud-based communication platforms. Bayesian probability models the relationship between email content features and the likelihood of spam. Conditional probability enables accurate classification by handling uncertainty in textual patterns. This probabilistic framework enhances filtering efficiency, reduces false positives, and improves adaptive learning in modern email security systems.
3	JOSHUA S URK25CO1001 SYED SAIFUDEEN URK25CO2002 ELAVARASAN R URK25CO2003 SUSHANTH ABHISHEK URK25CO2006	**Micro Project - 03**	Tumour Classification System Using Naive Bayes Theorem.
		Industry Focus	Healthcare Analytics & Intelligent Medical Systems.
		Integrated COs	CO1, CO3, CO5
		Project Scope	This project has strong future scope in real-time medical diagnosis and clinical decision support systems used in hospitals, diagnostic laboratories, and healthcare technology companies such as GE HealthCare, Siemens Healthineers, and Philips Healthcare. By applying the Naive Bayes Theorem, the system performs probabilistic classification of tumours as benign or malignant using clinical features such as cell size, texture, radius, and smoothness. This approach enables intelligent medical decision-making under uncertainty, supporting early cancer detection and predictive healthcare analytics. The system can be further extended to AI-based radiology support systems for automated tumour detection from medical images, histopathological image classification for precise tissue analysis, personalized treatment recommendation systems based on patient-specific parameters, and autonomous diagnostic platforms in smart hospitals that facilitate real-time, data-driven medical decisions. Overall, the model enhances diagnostic accuracy, reduces errors, and assists healthcare professionals in evidence-based treatment planning.

4	SWARNADEEP DEY URK25CO1004 SHAWN S URK25CO1006 REGAN JESURAJU URK25CO1010 JEEVAN JOYIT URK25CO2004	**Micro Project - 04**	User Behaviour Modelling Using Random Variables and Probability Density Functions
		Industry Focus	User Analytics & Recommendation Systems
		Integrated COs	CO1,CO2,CO4
		Project Scope	The project enables behavioural analytics for companies like Netflix, Amazon, and Spotify. Random variables and PDFs help model user engagement and interaction patterns. This supports personalized recommendations and performance optimization. The approach is valuable for data-driven decision systems in large-scale platforms.
5	Gokulnath.G URK25AI1129 Harish.p URK25AI1152 Tharun kumar.N URK25AI1136 Rohithsaran.U URK25AI1134 Jabaslin.J URK25AI1145	**Micro Project - 05**	Feature Relationship Analysis in Machine Learning Pipelines Using Joint Probability Distributions
		Industry Focus	Machine Learning Platforms
		Integrated COs	CO2, CO4,CO5
		Project Scope	This project has strong scope in improving ML model interpretability for companies like Google AI, OpenAI, and NVIDIA. Joint probability distributions help quantify dependencies between features. This improves feature engineering and reduces redundancy. It supports scalable and explainable AI systems.
6	Sajrin.R URK25AI1146 Sabeena K URK25AI1144 Madhumitha S URK25AI1128 Ritharsana V URK25AI1139	**Micro Project - 06**	Communication Network Traffic Burst Detection Using Poisson Distribution
		Industry Focus	Telecommunications & Networking
		Integrated COs	CO3, CO4, CO5
		Project Scope	The project supports traffic modelling solutions for companies like Ericsson, Nokia, and Cisco. Poisson distribution enables accurate modelling of packet arrivals. This helps detect abnormal traffic bursts and congestion. It can be extended to real-time network monitoring tools.
7	Vikas S (URK25AI1148) Dhiraj Kumar S (URK25AI1155) Jeeva V (URK25CO2001) Jeraldu L (URK25CO2005)	**Micro Project - 07**	Packet Loss Estimation in Data Networks Using Binomial Distribution
		Industry Focus	Network Performance Engineering
		Integrated COs	CO3-CO5
		Project Scope	This project can be applied in network diagnostics platforms used by ISPs and cloud providers like AWS and Azure. Binomial distribution models packet success and failure events. This enables early prediction of network degradation. It supports proactive network management solutions.
8	NAVEEN - (URK25AI1130) NITIEESHWAR S -(URK25AI1131) Stephan J - (URK25AI1125) A M ANTONY JONES SELVAN - (URK25AI1124)	**Micro Project - 08**	Server Response Time Modelling Using Exponential Distribution
		Industry Focus	Cloud Computing & Web Services
		Integrated COs	CO3-CO5
		Project Scope	The project has strong relevance for cloud service providers such as Amazon Web Services and Google Cloud. Exponential distribution helps model service times and delays. This supports load optimization and SLA monitoring. It can be integrated into performance analytics dashboards.
9	Dhiya S -	**Micro Project - 09**	Traffic Volume Modelling in Data Networks Using Normal Distribution

	URK25AI1140 Shanmitha M S - URK25AI1133 Ashna Anand - URK25AI1150 ALBIN JARVIS A URK25AI1143	**Industry Focus**	Distributed Systems
		Integrated COs	CO3-CO5
		Project Scope	This project models traffic volume in data networks using the Normal distribution to understand regular usage patterns. It involves collecting traffic data and calculating the average and variation of network load. The Normal model helps identify whether traffic changes are normal or unusual. The system checks for sudden deviations that may indicate congestion or abnormal activity. The project shows how statistical analysis supports efficient network monitoring and performance management..
10	Paulin Mercy URK25CO1007 Siva Priya URK25CO2009 Stephen Paul URK25CO18 Sumithra M URK25AI1127	**Micro Project - 10**	System Performance Benchmarking Using Measures of Central Tendency
		Industry Focus	Software Performance Engineering
		Integrated COs	CO2-CO5
		Project Scope	The project supports benchmarking tools for software platforms used by enterprises. Mean, median, and mode help summarize performance metrics. This enables comparative analysis of system efficiency. It can be extended to automated performance reporting tools.
11	Rose infanta . A URK25CO2011 Yuthika.G URK25CO2012 Dharunya.S URK25CO2017 karunya URK25CO2010	**Micro Project - 11**	Cloud Resource Utilization Prediction Using Moving Average Method
		Industry Focus	Operating Systems & Cloud Infrastructure
		Integrated COs	CO3-CO5
		Project Scope	This project focuses on predicting cloud resource utilization using the Moving Average method for time series analysis. It involves collecting historical data on CPU, memory, or storage usage from cloud systems. The Moving Average technique is applied to smooth fluctuations and identify usage trends over time. The model forecasts future resource demand to support efficient allocation and capacity planning. The project demonstrates how time series methods improve performance monitoring and optimize cloud infrastructure management.
12	SAM VIJAY S - URK25AI1149 VISHAL - URK25AI1126 ADHARSH - URK25AI1132 SHAM SANGTHUS - URK25AI1138	**Micro Project - 12**	System Bottleneck Identification Using Correlation and Rank Correlation
		Industry Focus	Software Analytics
		Integrated COs	CO2-CO5
		Project Scope	The project helps identify performance bottlenecks in complex software systems. Correlation techniques quantify relationships between system metrics. This improves debugging and optimization. It is valuable for DevOps and performance monitoring tools.
13	Jawahar Jaswanth URK25AI1142 Sujai URK25CO1005	**Micro Project - 13**	Server Response Time Prediction Using Regression Models
		Industry Focus	Web Application Engineering
		Integrated COs	CO2-CO5
		Project Scope	This project supports predictive analytics for web services. Regression models estimate response time based on workload parameters. This

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<tr><td></td><td>DAVE V SHAH URK25CO2008 VIAN ALEN BARNES URK25CO2019</td><td><table><tr><td></td><td>improves QoS and user experience. It can be integrated into intelligent monitoring systems.</td></tr></table></td></tr>
<tr><td>14</td><td>Vetri-URK25AI1147 Sanjay-URK25AI1153 Pinesh-URK25AI1154 Joshua-URK25AI1135</td><td><table><tr><td>Micro Project - 14</td><td>Server Performance Anomaly Detection Using Student's t-Test</td></tr><tr><td>Industry Focus</td><td>Intelligent Monitoring Systems</td></tr><tr><td>Integrated COs</td><td>CO1, CO4, CO5</td></tr><tr><td>Project Scope</td><td>This project applies the student's t-test to monitor server performance and detect anomalies in real time. It involves collecting historical server data to establish a normal performance baseline and continuously comparing new data against this baseline. The system identifies statistically significant deviations that may indicate faults, overload conditions, or cyber threats. The project can be integrated into AI-driven monitoring platforms to enhance automated decision-making and improve infrastructure reliability.</td></tr></table></td></tr>
<tr><td>15</td><td>Kidhiyon prabu URK25CO1009 Nanthini URK25CO1002 Jenita Reshma URK25CO1008 ABEL BATOI URK25CO101</td><td><table><tr><td>Micro Project - 15</td><td>Feature Independence Analysis in Data Science Using Chi-Square Test.</td></tr><tr><td>Industry Focus</td><td>Statistical Feature Engineering & Data Quality Analytics</td></tr><tr><td>Integrated COs</td><td>CO2, CO4, CO5</td></tr><tr><td>Project Scope</td><td>This project studies whether categorical features in a dataset are independent using the Chi-Square test. It involves selecting suitable data and applying the test to check relationships between variables. The results help in choosing important features for building predictive models. The project shows how statistical testing improves data analysis and model performance.</td></tr></table></td></tr>
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